

Fishing for biodiversity by balanced harvesting

A mizer reimplementation of Law & Plank (2023), *Fish and Fisheries*

A reimplementation in [mizer](#) of

Law, R. & Plank, M.J. (2023) *Fishing for biodiversity by balanced harvesting*. **Fish and Fisheries** 24, 1–17. doi:[10.1111/faf.12705](https://doi.org/10.1111/faf.12705) (preprint: doi:[10.1101/2021.06.27.450047](https://doi.org/10.1101/2021.06.27.450047))

Source: github.com/gustavdelius/balanced-harvest

The question

Can fishing, if scaled sensibly across species, help maintain biodiversity rather than erode it? Law & Plank compare three ways of setting the fishing mortality rate of species i :

	rule	
fixed	$F_i(t) = F$	Eq. (2.7)
BH _P	$F_i(t) = c_P P_i(t)$	Eq. (2.8)
BH _{P/B}	$F_i(t) = c_{P/B} P_i(t) / B_i(t)$	Eq. (2.9)

Every fish enters a single mixed-species fishery at $w_f = 400$ g, and everything above that is caught at the same rate, so F is size-independent and the balancing is **across species, not across sizes**. B_i and P_i are measured over that same harvested range, and $c_{P/B}$ is simply a fixed exploitation ratio $E = Y_i/P_i$ applied to every species.

The argument is that because biomass and production rate are tightly coupled across species,

$$B_i \sim P_i^\alpha, \quad \alpha \approx 1,$$

setting $F_i \propto P_i$ makes fishing mortality roughly proportional to biomass, so it falls away as a stock falls. Fixed F has no such feedback, and P_i/B_i barely changes as a stock thins, so BH_{P/B} has almost none either.

α is the regression slope of $\log B_i$ on $\log P_i$ across the species of an assemblage, and it recurs throughout this site: it is what makes BH_P a feedback rule rather than a fixed allocation, and Figure 4 below is a direct test of it.

The model, in equations

The quickest way to say what Law & Plank's model is, is to write it down. Everything here is Appendix A of the paper, restated in mizer's variables: the per-mass number density $N_i(w, t)$ of species i , where the paper uses the log-density $u_i(x, t) = w N_i$ with $x = \ln(w/1\text{ g})$. The two are the same PDE divided through by w ; [mapping.md](#) does that term by term.

The dynamics (Eq. A.1) are McKendrick-von Foerster with a diffusion term:

$$\frac{\partial N_i}{\partial t} = -\frac{\partial}{\partial w}[g_i(w, t) N_i] + \frac{1}{2} \frac{\partial^2}{\partial w^2}[D_i(w, t) N_i] - \mu_i(w, t) N_i,$$

with eggs entering at $w_0 = 10^{-3}$ g at the rate $R_i(t)$ given below. The diffusion term is there because eating a prey item is a discrete jump in mass: D_i is the second moment of the same prey integral whose first moment gives g_i .

Feeding (Eqs A.2, A.4). Search rate is $A_i w^{\alpha_q}$, and the feeding kernel is a **box** spanning four decades of predator-prey mass ratio, $e^{\beta \pm 3\sigma} = 10$ to 10^5 , with height $1/(6\sigma)$:

$$E_i(w, t) = \frac{A_i w^{\alpha_q}}{6\sigma} \int_{w/10^5}^{w/10} [N_R(w_p, t) + \sum_j \theta_{ij} N_j(w_p, t)] w_p dw_p.$$

There is **no satiation and no metabolic cost**, so the whole assimilated intake is split between growth and reproduction by the allocation $\epsilon_i(w) = 1 - \psi_i(w)$ of Eq. (A.8):

$$g_i(w, t) = \epsilon_i(w) K E_i(w, t), \quad \psi_i(w) = \frac{1}{1 + (w/w_{\text{mat}, i})^{-\rho_m}} \left(\frac{w}{w_{\text{max}, i}} \right)^{\rho_\infty},$$

with $w_{\text{mat}, i} = w_{\text{max}, i}/10$. Eggs are produced without any stock-recruitment relationship (Eq. A.9):

$$R_i(t) = \frac{\epsilon_R}{w_0} \int \psi_i(w) K E_i(w, t) N_i(w, t) dw.$$

Mortality is predation, a larval term and a background term, $\mu_i = \mu_i^{\text{pred}} + \mu_i^L + \mu_i^b$. Predation (Eq. A.3) is the mirror of the encounter integral,

$$\mu_i^{\text{pred}}(w_p, t) = \sum_j \theta_{ji} \frac{A_j}{6\sigma} \int_{10w_p}^{10^5 w_p} w^{\alpha_q} N_j(w, t) dw,$$

the larval term (Eq. A.6) is a reverse sigmoid that has died away by $w_L = 0.1$ g,

$$\mu_i^L(w) = \frac{\mu_{\text{egg}, i}}{1 + (w/w_L)^{\rho_L}},$$

and the background term (Eq. A.7) is **food-dependent** — metabolic need relative to how fast food is arriving, so it is a starvation feedback rather than a constant:

$$\mu_i^b(w, t) = \mu_b^{(0)} \left(\frac{w}{w_0} \right)^{-\xi} \frac{\tilde{g}_i(w_0, t)}{\tilde{g}_i(w, t)}, \quad \tilde{g}_i = K E_i/w.$$

That last ratio is printed the other way up in the paper, which changes the model qualitatively; [mu_b.md](#) sets out why this implementation inverts it.

The plankton (Eqs A.11-A.14) is logistic with immigration, and has no size structure of its own — cells neither grow nor are born into a smaller class:

$$\frac{\partial N_R}{\partial t} = r(w) N_R \left(1 - \frac{N_R}{a(w)}\right) - \mu_R^{\text{pred}}(w, t) N_R + I(w),$$

with $r(w) = r_0 w^{-\rho}$, and carrying capacity a and immigration I both power laws in w anchored at 1 mg with $I_0 = a_0$ — so $I/a = 1 \text{ yr}^{-1}$ at *every* plankton size, which turns out to matter a great deal — see [Is the result robust?](#) below.

What is harvested (Eqs 2.3, 2.4, 2.6). Everything above the entry mass w_f is fished at the same rate, and biomass and production are measured over that same range:

$$B_i(t) = \int_{w_f}^{w_{\max,i}} w N_i(w, t) dw, \quad Y_i(t) = F_i(t) B_i(t),$$

$$P_i(t) = \underbrace{\int_{w_f}^{w_{\max,i}} g_i(w, t) N_i(w, t) dw}_{\text{growth inside the range}} + \underbrace{[w g_i N_i]_{w_f}}_{\text{flux in}} - \underbrace{[w g_i N_i]_{w_{\max,i}}}_{\text{flux out}}.$$

P_i is **somatic production** — the rate at which the harvested part of the stock puts on flesh, plus the biomass that grows across w_f into the range. It is not yield, and not surplus production in the stock-assessment sense. The flux out of the top vanishes, because all food goes to reproduction at $w_{\max}$.

The parameters are the paper's Tables 1 and 2, reproduced verbatim in [R/lp_constants.R](#):

A_i	$37.5 \text{ m}^2 \text{ yr}^{-1} \text{ g}^{-\alpha_q}$	search-rate coefficient
α_q	0.85	search-rate exponent (mizer's q)
β, σ	6.908, 1.535	centre and width of the box kernel, in log PPMR
K	0.1	conversion efficiency (mizer's α)
$\mu_b^{(0)}, \xi$	$0.1 \text{ yr}^{-1}, 0.15$	background mortality at egg size, and its size scaling
w_L, ρ_L	0.1 g, 5	where larval mortality dies away, and how sharply
ρ_m, ρ_∞	15, 0.2	sharpness of maturation, approach to $w_{\max}$
ϵ_R	0.2	reproductive efficiency
θ	0.5 / 0.2 / 1	cannibalism / between species / on plankton (Appendix C)
r_0, ρ	10 yr^{-1} at 1 g, 0.15	plankton rate of increase and its size scaling

$a_0 = I_0$	2000 m ⁻² at 1 mg	plankton carrying capacity and immigration
λ	2	slope of the isolated plankton spectrum
w_f	400 g	entry mass of the fishery
$\Delta x, \Delta t$	0.1, 0.002 yr	log-mass grid step, time step

Species differ in exactly two things: maximum body mass $w_{\max, i}$ and larval death rate $\mu_{\text{egg}, i}$. Everything else is shared across species. That single fact is what [life-history.md](#) is about.

Is this really the paper’s model?

The paper’s model is **not** a standard mizer model. It has no satiation and no metabolic cost, no imposed stock-recruitment relationship, food-dependent intrinsic mortality, logistic plankton with immigration instead of a semichemostat resource, and a box feeding kernel spanning four decades of prey size. All of that is built from Appendices A–C; the term-by-term correspondence is in [mapping.md](#).

Rather than assert the mapping, [tests/test_model.R](#) checks mizer’s rates against hand-coded integrations of the paper’s own equations on the model’s own grid. All sixteen checks pass — the encounter rate against Eq. (A.2) and predation mortality against Eq. (A.3) agree to about 1e-13, and the production integral of Eq. (2.4), including its boundary terms, converges at the expected first order.

Three independent quantities then came out close to the paper’s without being fitted to them:

	this reimplementaion	Law & Plank
α in $B \sim P^\alpha$ (four ecosystems)	0.977, 0.998, 0.993, 0.984	1.004
c_P calibrated against $F = 0.1$	0.906 m ² g ⁻¹	1
$c_{P/B}$ calibrated against $F = 0.1$	0.259	0.25

α matters most: it is the exponent the whole Figure 4 argument rests on (“species were near a line of slope $1 + \alpha = 2.004$ ”), it is set by how much adult mortality the model has, and it emerges here from four independently assembled ecosystems.

The ecosystems

There is no species pool taken from nature here. Following Appendix B, each ecosystem is **assembled inside the model**: introduce one fish species at a time, with maximum body mass drawn log-uniform over 100 g – 40 kg and larval death rate μ_{egg} drawn uniform over 28–32 yr⁻¹, start it at a low density on a power law, let the system relax for 50 years, cull anything below 2×10^{-6} g m⁻², and repeat until 15 species coexist. An assemblage is accepted only if it meets Appendix B’s four criteria: the species coexist, the state is close to equilibrium, $w_{\max}$ spans a wide range inside the drawn limits, and biomass spans roughly

four orders of magnitude. [tests/test_ecosystem.R](#) checks all four, plus the numbers the paper quotes in Sections 2.4 and 3.

Four ecosystems are used throughout this site, and their names recur on every page:

- **eco1** — seed 101, every species given the same search-rate coefficient A_i . This is the one Figures 2, 3 and 4 are drawn from, and the one most of the experiments run on.
- **eco_r1**, **eco_r2**, **eco_r3** — seeds 201–203, with A_i randomised per species ($A_i = z_i A$, $z_i \sim N(1, 0.1)$), which is what the paper does for its Figures 5 and 6. The extra life-history variation is the only difference.

Later pages add more, all named after the seed they were assembled from: **eco301** – **eco312** are twelve further ecosystems assembled under exactly the **eco1** protocol, used for the replication in [robustness-results.md](#); and the **varied_***, **fixed_***, **neutral_*** and **gradient_*** runs of [life-history.md](#) are named *experimental arm* underscore *seed*.

	eco1	eco_r1	eco_r2	eco_r3	paper
species	15	15	15	15	15
invasion attempts	23	25	30	33	25, 24, 19, 45
$w_{\max}$ range	327 g – 13.6 kg	860 g – 9.3 kg	459 g – 14.5 kg	566 g – 22.3 kg	100 g – 40 kg
biomass span	5.8 decades	3.0	3.8	7.8	~4
cor(log B, log P)	0.999	0.998	0.997	1.000	—
primary production (g m ⁻² yr ⁻¹)	4310	4332	4324	4312	~4000
total fish biomass (g m ⁻²)	5.95	6.24	5.82	5.51	~4–5
yield at $F = 0.1$ (g m ⁻² yr ⁻¹)	0.287	0.503	0.416	0.289	~0.25
Fogarty ratio (‰)	0.067	0.116	0.096	0.067	0.06

The paper does not report its random seed or the realised life histories of its 15 species, so this reproduces the assembly *procedure*, not the particular assemblage. The figures below are therefore comparable in kind, not number for number.

Figure 2 — the biomass-production relationship

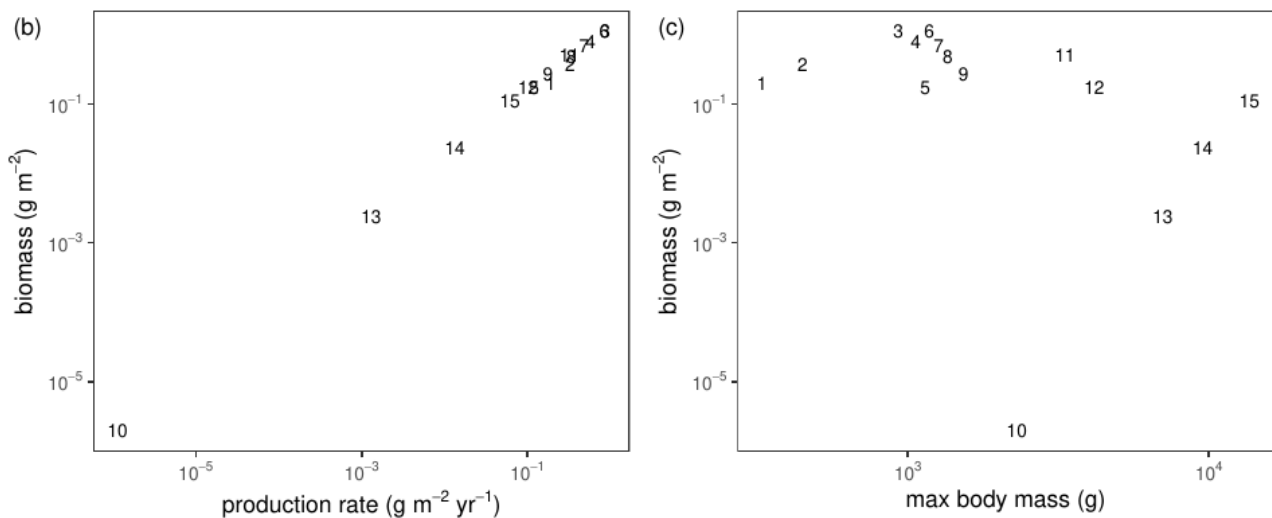


Figure 2

Panel (b) is the relationship the whole argument depends on: biomass and production rate are tightly correlated across species, over six decades. Panel (c) shows the pattern the paper reports in its own Fig. 2c — the small- $w_{\max}$ species are the common ones, and the rare species sit at larger maximum body mass.

Why panel (b) comes out this way, and what it takes to break it, is worked out in [life-history.md](#): P/B is the biomass-weighted mean mortality rate, and mortality here is a function of body size that no species can escape, so abundance cancels exactly. Two attempts to spread P/B across species are reported there. Giving every species its own egg mass fails — assembly drives all survivors to the largest egg on offer. Giving them their own activity succeeds, but only once faster species are made correspondingly more vulnerable to predation, which turns activity into a neutral axis that assembly cannot climb; a genuine fast-slow continuum then assembles, with a ten-fold range in age at maturity. $B \sim P$ survives even that.

Running the three rules on those ecosystems is also the sharpest test of the paper's conclusion so far. $BH_{P/B}$ is no longer a near-duplicate of a fixed F there — it spans an order of magnitude and tracks turnover at $r = 0.94$, which is orthodox single-stock advice — and it *still* leaves the rare species exposed, because turnover is orthogonal to rarity. Only tracking abundance protects biodiversity.

(The paper's Fig. 2a is a scatter from an Ecopath model of the West Scotland shelf. That is empirical data, not model output, and is not reproduced here.)

Figure 3 — three ways to harvest the same assemblage

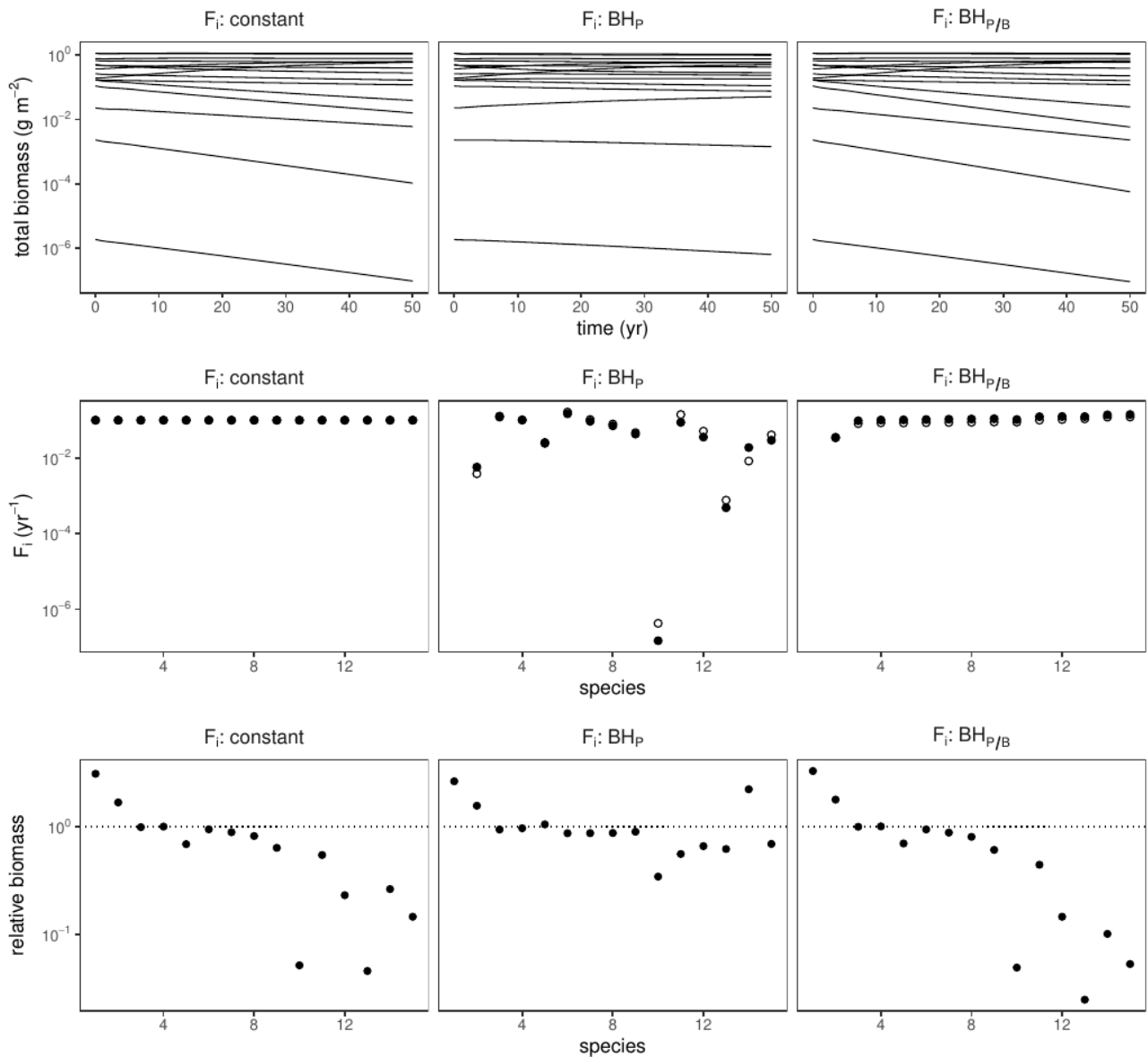


Figure 3

All three regimes are calibrated to the same total yield after 50 years ($0.197\text{--}0.198 \text{ g m}^{-2} \text{ yr}^{-1}$).

Top row, total biomass of each species over the 50 years of fishing. Under fixed F and $BH_{p/B}$ several species fall along straight lines in log biomass — exponential decline. Under BH_p the trajectories are comparatively flat.

Middle row is the mechanism, laid bare. Fixed F is flat across species by construction. $BH_{p/B}$ is *also nearly flat* — which is the paper's point: P_i/B_i is much the same for every species, so a constant exploitation ratio is barely distinguishable from a constant F . BH_p spans seven orders of magnitude, dropping to $F \approx 10^{-7} \text{ yr}^{-1}$ for the rarest species.

Bottom row, the ratio of year-50 to year-0 biomass. Under fixed F and $BH_{P/B}$ the larger-numbered (rarer) species fall well below the dotted line; under BH_P they stay close to it.

Figure 4 — yield against production rate

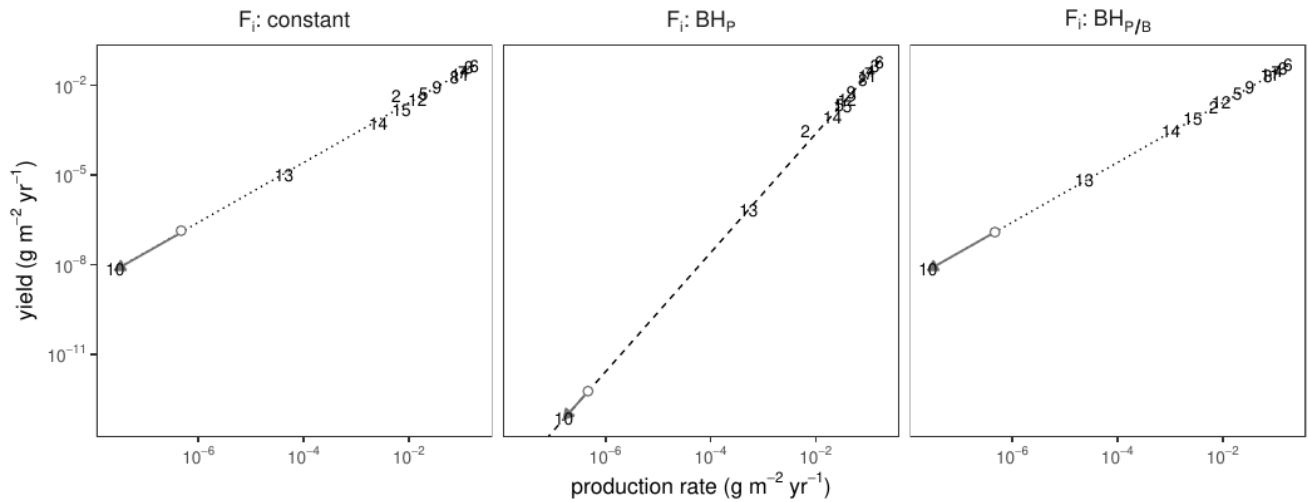


Figure 4

Dotted lines are isoclines of constant exploitation ratio E ; the dashed line in the middle panel has slope $1 + \alpha$.

Under fixed F and $BH_{P/B}$ the species lie along a line of slope 1 — every species is exploited at the same ratio, regardless of how rare it is. That is what puts the rare ones on a downward path. Under BH_P the species lie along the steeper slope- $(1 + \alpha)$ line: yield rises faster than linearly with production, so rare species are exploited at a progressively *smaller* fraction of their production. The open circle and arrow trace the rarest fished species from year 0 to year 50.

This is the paper's central quantitative claim, and it reproduces.

Figure 5 — emergent growth

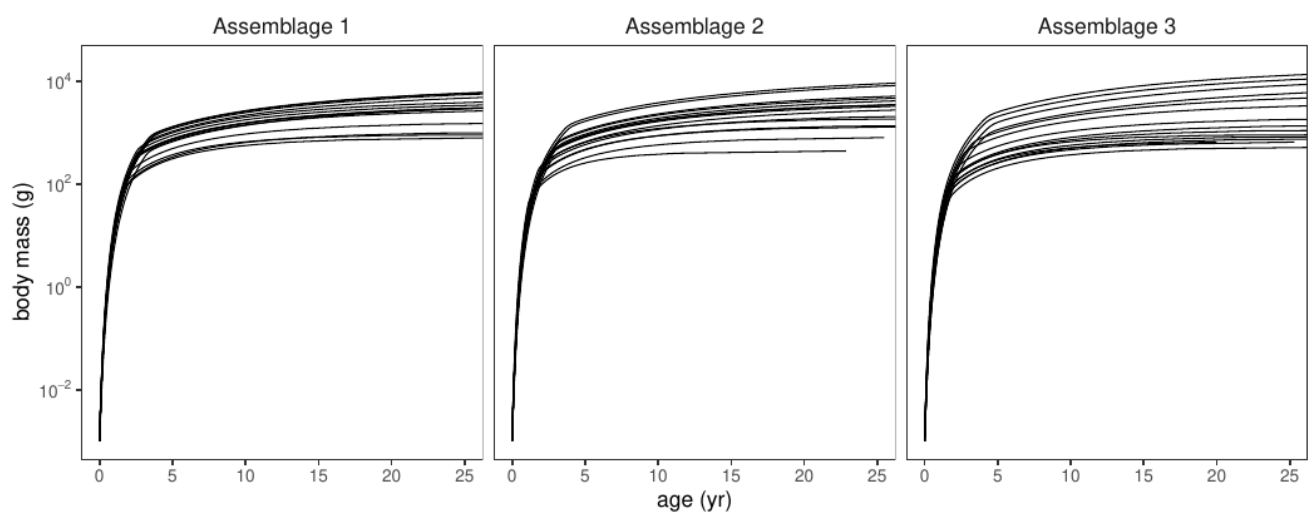


Figure 5

Growth trajectories obtained by integrating Eq. (C.1). Nothing here is imposed: body growth emerges from what the fish actually eat. By age 1 the species span a 3.3- to 3.7-fold range in body mass, against the 4- to 7-fold range the paper reports for its own assemblages.

Figure 6 — three more ecosystems, harder fishing

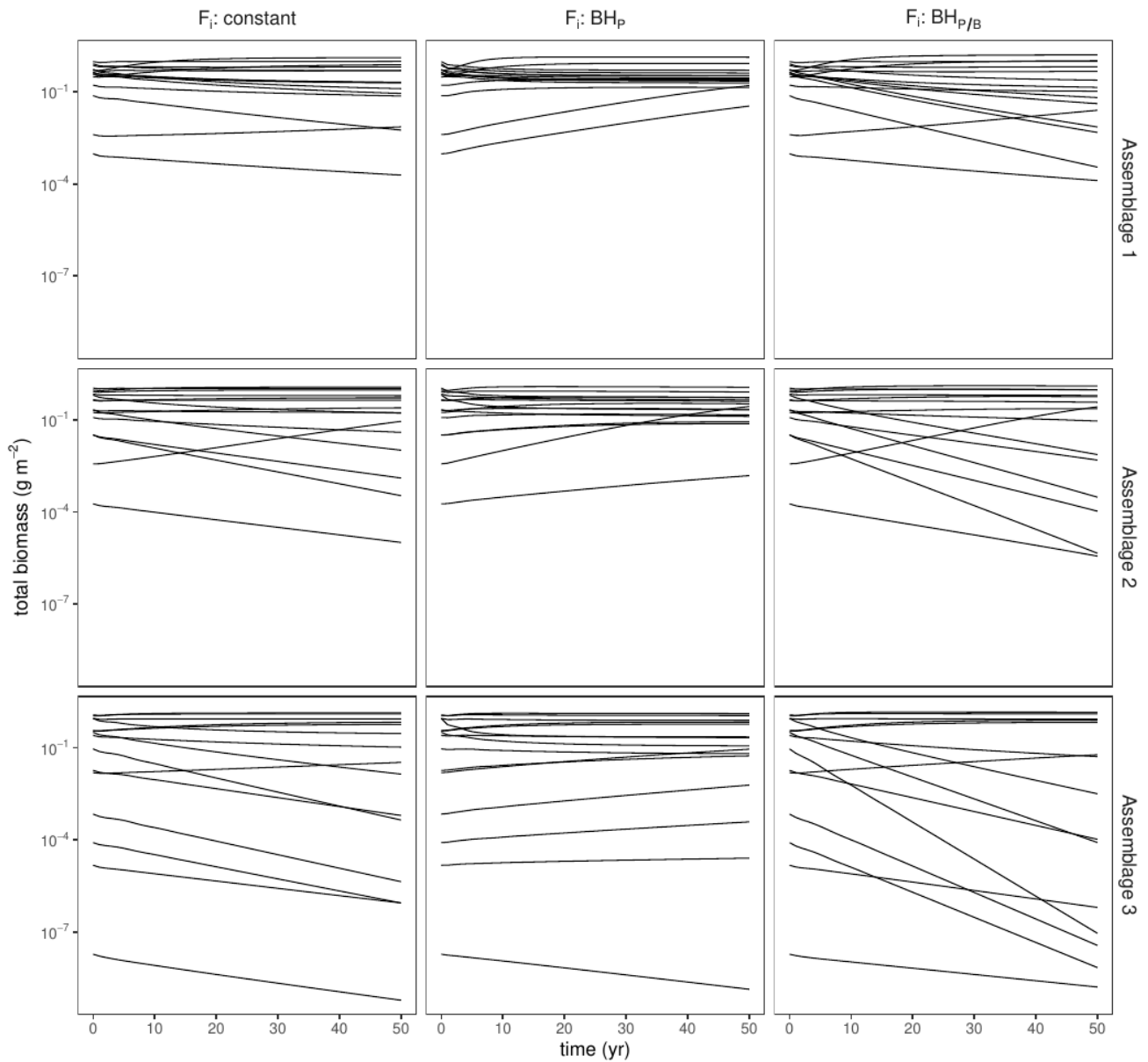


Figure 6

Three independently assembled ecosystems (rows) with extra life-history variation, baseline fishing doubled to 0.2 yr⁻¹, and a random per-species intensity factor $z'_i \sim U(0.5, 1.5)$ applied identically to all three rules. The BH_p column is visibly flatter than either neighbour in all three assemblages, and the BH_{p/B} collapses in assemblage 3 are dramatic.

The headline comparison

On `eco1`, with all three regimes calibrated to the same total yield:

regime	yield (g m ⁻² yr ⁻¹)	worst species, vs year 0	worst species, vs unfished control	species below 10% of control
unfished control	0	0.174	1.00	0
fixed <i>F</i>	0.198	0.046	0.166	0
BH_P	0.198	0.345	0.573	0
BH _{P/B}	0.197	0.025	0.070	3

A caveat on the “worst species” columns, found while running these rules on the fast-slow ecosystems of [life-history.md](#): under BH_P the worst-affected species is usually the *most abundant* one, because that is where the rule deliberately concentrates effort. The columns therefore measure harm to a common species under one rule and harm to a rare one under another. Conditioning on rarity instead changes BH_P’s measured advantage there from 0.95-1.6x to 5.4-16.9x.

Across all four ecosystems, counting species that end below 10% and below 1% of what they would have been with no fishing at all:

	eco1	eco_r1	eco_r2	eco_r3
fixed <i>F</i>	0 / 0	0 / 0	3 / 0	5 / 2
BH_P	0 / 0	0 / 0	0 / 0	0 / 0
BH _{P/B}	3 / 0	3 / 1	6 / 3	6 / 6

BH_P does not push a single species below 10% of its unfished trajectory in any of the four ecosystems, and its worst-affected species retains 25–57% of unfished biomass. In the three Figure 6 ecosystems it does this while returning *more* yield than fixed *F* (0.797 vs 0.631, 0.539 vs 0.422, 0.369 vs 0.344), so it is not buying protection by fishing less.

The paper’s conclusion reproduces.

One correction to how the comparison is framed

Law & Plank compare year-50 fished biomass against year-0 unfished biomass. But the assembled ecosystem is explicitly only a quasi-equilibrium — the paper says so — and it drifts. In the unfished control run here, the worst-affected species still falls to **17% of its year-0 biomass over 50 years with no fishing at all**, and individual species move by 39–86% over a further 20 unfished years.

Comparing against year 0 therefore charges fishing for drift it did not cause. The effect is not small: fixed *F* on eco1 looks like it takes its worst species to 4.6% of where it started, but only to 17% of where it would have been anyway, and the count of species below 10% falls from 2 to 0 once the control is used. The ranking of the three rules is unchanged, but the magnitudes are not. Both columns are reported above.

Where this departs from the paper

Two places in the paper needed a judgement call, and one of them matters a great deal. Full detail in [mapping.md](#) and [mu_b.md](#).

Eq. (A.7) contradicts itself. The prose says background mortality is proportional to metabolic need *divided* by the rate food becomes available; the printed formula *multiplies* by it. The two differ by two orders of magnitude in adult mortality. The printed version gives near-zero adult mortality, biomass piling up at $w_{\max}$, roughly ten times too much fish biomass, and large species so dominant that small ones cannot invade at all. The prose version reproduces the paper's own reported biomass range, its $0.25 \text{ g m}^{-2} \text{ yr}^{-1}$ yield, its 0.06 ‰ Fogarty ratio, and its ordering in which small species are common and large ones rare. This implementation follows the prose; `mu_b_form = "product"` gives the other.

Three smaller ones: Eq. (A.3) as printed evaluates the predator's search area at the *prey's* body mass (Eq. A.2 uses the consumer's own size, so this is a typo); Table 1 contradicts the text on the plankton carrying-capacity offset (the text is right, and gives the stated $4000 \text{ g m}^{-2} \text{ yr}^{-1}$ of primary production); and Appendix B does not say what power law an invading spectrum starts on.

Is the result robust?

Reproducing a result is not the same as testing it. [robustness.md](#) works through which of the paper's arbitrary choices could change the conclusion — the year-50 yield-matching criterion, the absence of compensatory recruitment, whether $F \propto B$ would do just as well as $F \propto P$, and whether the *adaptive feedback* or merely the *initial allocation* is doing the work — and sets out an investigation.

[robustness-results.md](#) reports what came out of it — phases 1, 2, 4 and 5 and Phase 3 items 8, 8b, 10 and 11, across up to thirteen independently assembled ecosystems. In short: the calibration criterion turns out not to matter (the frontiers never cross, and I was wrong to call it the most load-bearing choice); F proportional to *biomass* does everything F proportional to production does; the adaptive feedback turns out to be essential, though it took thirteen ecosystems to establish that — on one it looked as though the initial allocation did most of the work; the plankton cannot be made contestable without destroying the large-bodied community the paper harvests; and, most consequentially, imposing a stock-recruitment relationship on the published ecosystem while holding the community exactly fixed shrinks BHP's advantage over fixed F from $3.4\times$ to $1.0\times$. Replicated over thirteen independently assembled ecosystems, the ranking $\text{BHP} > \text{fixed} > \text{BHP}_B$ holds 13 times out of 13, with the advantage varying fourfold (median $2.4\times$). The feedback tolerates being updated only once a decade, but not observation error much beyond a factor of two.

One finding from that work is already in hand. Appendix B nominates competition among larvae for plankton as the mechanism that replaces an imposed stock-recruitment relationship. Measured on the assembled 15-species ecosystem, the plankton over the entire larval prey range (10^{-8} to 10^{-4} g) sits **at its carrying capacity**, and doubling the whole fish spectrum changes growth at egg size by only -1.3% (against -91% at 100 g).

The reason is structural: the paper sets immigration I_0 equal to carrying capacity a_0 with the same size scaling, so $I/a = 1$ per year at every plankton size, while plankton at larval prey sizes regenerate at $40\text{--}220\text{ yr}^{-1}$. The model therefore has neither an imposed recruitment brake nor, at these parameter values, the emergent one offered in its place.

Reproducing this

```
Rscript tests/test_model.R      # verify the mapping; needs no inputs
Rscript run_assembly.R          # ~1 h -> data/ecosystems.rds
Rscript tests/test_ecosystem.R  # check the ecosystems against Appendix B
Rscript run_figures.R           # ~40 m -> data/results.rds, figures/
Rscript tests/test_convergence.R # the numerical liberties taken
```

The life-history experiments of [life-history.md](#) are separate and slower, one assembly per invocation:

```
Rscript run_egg_assembly.R 301 varied      # -> data/egg/
Rscript run_activity_assembly.R 301 neutral # -> data/activity/
Rscript run_activity_harvest.R 301         # -> data/activity_harvest/
```

Requires [mizer](#) (developed against 3.3.0.9000), ggplot2 and patchwork.

Where does $B \sim P$ come from?

[life-history.md](#) works out why production is proportional to biomass in these ecosystems, and tries twice to break the relationship by giving species life histories that vary independently of maximum body mass.

A second test, on a real ecosystem

The ecosystems above are assembled inside the model, which is the right way to test the paper but says nothing about whether the result survives contact with a real one. [north-sea.md](#) runs the same three rules on mizer's North Sea model with real depleted species — thornback ray and Atlantic halibut — added at abundances estimated from ICES IBTS survey data.

The conclusion holds there too, but conditionally: `NS_params` as it ships has nothing capable of being lost, and BH_P looks *worst* of the three until something at risk is put in. Halibut then goes from 2% of its unfished trajectory under a constant F to 540% under BH_P , at equal yield, while $BH_{P/B}$ destroys the thornback ray.

Caveats

The ecosystems are not the paper's ecosystems — the assembly procedure is reproduced, not the particular assemblage, because the seed and realised life histories are not reported. Four ecosystems is a small sample for a claim about assemblages in general. And the comparison rests on a single calibration criterion; the trade-off frontier proposed in [robustness.md](#) would be a stronger design than the one used here and in the paper.

Page <https://gustavdelius.github.io/balanced-harvest/> of the site accompanying the mizer reimplementation of Law & Plank (2023). Generated by `make_pdfs.py`.